

Designing an Optics Simulation with Gravitational Lensing

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Abstract—This report discusses the design of a 3D optics simulation with the capability to simulate gravitational lensing. The deflection angle of light from a gravitational lens was tested against the theoretical formula, and the simulation appears to underestimate it by approximately a factor of two, likely due to limitations in the simulation.

I. INTRODUCTION

PHYSICISTS have come up with ways to describe how light behaves. We can treat it like a particle. We can treat it like a wave. However, simply knowing the laws that govern the behaviour of light is somewhat limited in that it does not give us a more visual understanding of how light actually behaves. In this report, I discuss the process of making an optics simulation following the guidance of ref. [1]. I will also investigate how gravitational lensing can be modelled with this simulation and to what extent it agrees with our current theoretical predictions.

This report goes through the techniques used in making this simulation and then investigate the angular bend in light that undergoes gravitational lensing according to the simulation.

II. DESIGNING THE SIMULATION

The simulation was structured into modules and classes in each module. This allowed for customisation of the optical setup. The main two types of objects in the simulation are rays and optical elements, which rays are propagated through.

1) *Lenses and Rays*: The rays are defined by a list of different points that the ray has travelled through, and the direction that the ray is currently facing. Then propagating the ray involves adding another point and updating the direction vector based on the specific interaction.

Then, the optical elements were mainly spherical surfaces which could either refract or reflect a ray according to Snell's Law and the laws governing reflection. Surfaces could be planar by instantiating a spherical surface with no curvature. These surfaces can be used to create planoconvex and biconvex lenses. For each interaction between a ray and a surface, the intersection point along with the updated direction vector are calculated. This is the main structure of the simulation.

2) *Gravitational Lensing*: So far, the simulation models light bending in discrete steps at surfaces while gravitational lensing is not like this. There is a continuous bend in the light as it approaches a heavy object. In order to simulate this, a gravitational lens had to be composed of many spherical shells where each shell deflects the ray by a small amount. The existing physics in the simulation was utilised by using a radially symmetric refractive index to mimic gravitational

lensing. The formula for refractive index (from ref.[2]) that does this is as follows:

$$n(r) \approx 1 - \frac{2\Phi(r)}{c^2}, \quad (1)$$

where $\Phi(r)$ is the radial gravitational potential of the object $-\frac{GM}{r}$. Substituting this in yields this equation for the weak field approximation.

$$n(r) \approx 1 + \frac{2GM}{c^2 r} \quad (2)$$

Each shell would have a refractive index of $n(r + \delta r)$ on the outside and $n(r)$ on the inside. The shells have linearly increasing radii up to a maximum radius of influence of a gravitational lens, so the δr is just the maximum radius of influence divided by the number of shells. With this implemented, the rays could be shown to bend around the gravitational lens, as shown in Fig. 1. However, problems arose when the rays got too close to the centre of the object, as the refractive index became negative. This caused rays to bend away from the object, which is unphysical. The point at which these problems occurred was actually the Schwarzschild radius of the object, so the decision was made to make the shell at the Schwarzschild radius opaque to stop these problems.

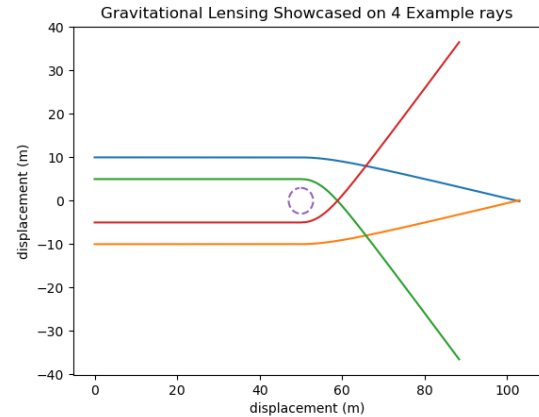


Fig. 1. 4 example rays being propagated through a gravitational lens. The dashed circle is the Schwarzschild radius of the gravitational lens. The mass of this particular lens was $2 * 10^{27}$ kg.

III. INVESTIGATING THE DEFLECTION ANGLE

The theoretical total angular deflection in light due to gravitational lensing is given by the following formula from Ref.[3]:

$$\Delta\theta = \frac{4GM}{c^2 r}, \quad (3)$$

where r is the distance of closest approach of the light ray if it did not bend.

The optical setup involves many rays of light that are initialised to all face the positive z direction and fairly far from the gravitational lens, but at different y values, as depicted in Fig. 2. They are then propagated through the gravitational lens and then the deflected angle is calculated from the dot product. The traced plot and the graphs of the calculated against the theoretical deflected angle are shown below.

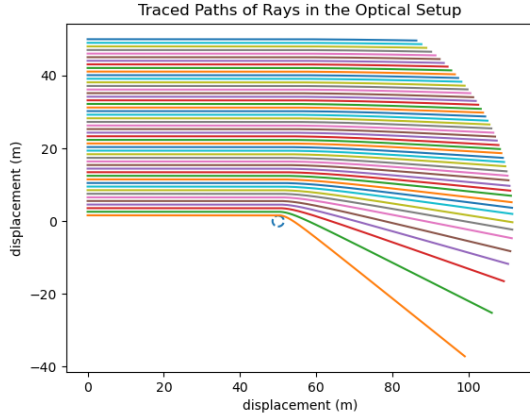


Fig. 2. The setup involving the rays being traced as they bend around the gravitational lens. There are 50 rays equally spaced by 1m each. In this plot, the rays are depicted as coming from much closer than they actually come from. Here, the lens is placed at $z = 50\text{m}$, when in the actual test, the lens is at $z = 1000\text{m}$ to ensure that the rays of light go through all the shells to make the total angular deflection more accurate. The mass of the lens is $1 \times 10^{27}\text{kg}$, and it is split into 5000 shells..

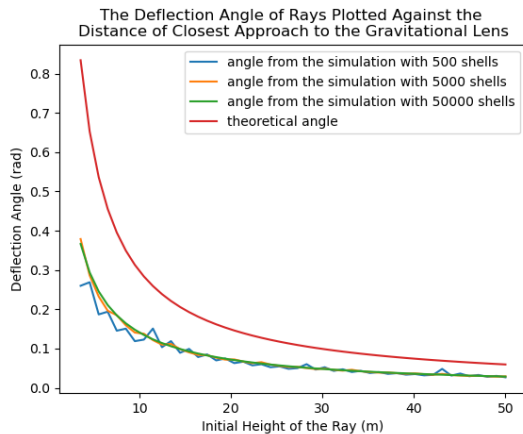


Fig. 3. This is the simulated angle of the deflection against the initial y value of the ray, which corresponds to distance of closest approach to the lens if there were no deflection. The number of shells is given in the graph and the maximum radius of influence of the lens is 200m. The mass of this lens is also $1 \times 10^{27}\text{kg}$.

The data suggests a consistent underestimate in the angle of deflection of the rays compared with the theoretical angle. There are multiple reasons for this, which potentially highlight

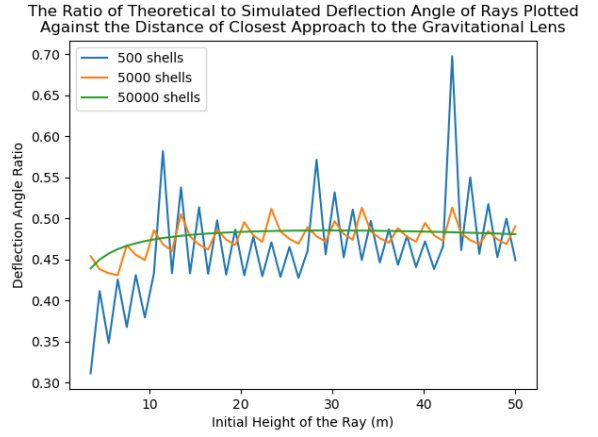


Fig. 4. This is the ratio of the simulated to theoretical deflection angle plotted against the distance of closest approach. For every resolution of simulation, the simulated angle is consistently lower than the theoretical angle, and as the number of shells increases, the value seems to approach a constant, which is around a half. The setup is the same as in Fig. 3

limitations in this simulation. One reason is the formula used for refractive index. This is a first order approximation and only applies for weak fields, however, at least the closer rays in the setup are likely close enough to the gravitational lens that this approximation is not valid. This could be causing the dip on the left hand side of Fig. 4. Another limitation is the finite radius of influence of the gravitational lens. As soon as the ray enters the radius of influence, it starts to bend and as soon as it leaves, it stops bending, when in reality, it will be continually bending throughout its journey as the radius of influence of a gravitational lens is infinite. Another potential source of error is the discrete nature of the system being broken down into shells, however, this is likely not a significant source of error, as we can observe the curves in Fig. 3 and Fig. 4 converging as we get to 50000 shells.

IV. CONCLUSION

The simulation manages to visually show the effects of gravitational lensing. It, however, is not able to accurately capture the angle that light is deflected by when encountering a gravitational body, as it significantly underestimates it by approximately a factor of two compared to theoretical predictions. This shows potential flaws and limitations in the methods used to capture this behaviour in the simulation. Future improvements could account for higher order terms in the refractive index, optimise a balance between number of shells and the maximum radius of influence, or potentially use a completely different system to determine the path of the rays.

REFERENCES

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